



Brainware University

Department of Computer Science & Engineering - Artificial Intelligence Final Year

Project Progress Report

Project Title: *"Detection of Skin Disease" A Placement Preparation App*

Group Members:

1. **Pradipta Maity**
Student Code: BWU/BTA/22/198
Roll No.: 22010332312
2. **Subrata Pan**
Student Code: BWU/BTA/22/232
Roll No.: 22010332211
3. **Argha Roy Khan**
Student Code: BWU/BTA/22/652
Roll No.: 23011332032
4. **Anirban Mondal**
Student Code: BWU/BTA/22/234
Roll No.: 22010332212

Project Guide: *Mr. Kaushik Paul, CSE-AI*

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Signatures:

Group Members:

1. _____(Signature with date)

2. _____(Signature with date)

3. _____(Signature with date)

4. _____(Signature with date)

Project Guide:

_____(Signature with date)

Date of Submission:

Project Report: Detection of Skin Disease

1. Introduction

Skin diseases are among the most common health problems affecting millions worldwide. Manual diagnosis is time-consuming and requires dermatologists' expertise. Deep Neural Networks (DNNs), particularly Convolutional Neural Networks (CNNs), provide an automated approach to detect and classify skin diseases early, reducing the burden on healthcare systems.

2. Project Overview

This project aims to build a deep learning-based system for skin disease detection using dermoscopic images. The HAM10000 dataset with over 10,000 images across 7 disease classes is used. Key Features:

- Image upload functionality.
- Automatic disease classification using CNN.
- Confidence score display.
- Basic skin-care tips based on detected disease.
- Option to view history and analysis charts.

3. Technologies Used

Frontend Stack:

- React.js – Component-based user interface.
- Tailwind CSS – Fast, responsive design.

Backend & ML Stack:

- Python with Flask/FastAPI – Lightweight backend APIs.
- TensorFlow/Keras – Deep learning model (CNN/Transfer Learning).
- Pandas & NumPy – Data preprocessing and analysis.
- SQLite/MongoDB – User history and prediction storage.
- REST API – Connects frontend and backend for predictions.

Machine Learning Features:

- Preprocessing: Resizing, normalization, augmentation.
- Model: MobileNetV2/ResNet50 for classification.
- Evaluation: Accuracy, precision, recall, F1-score.

4. Challenges Faced & Solutions

Challenges:

1. Dataset imbalance across 7 skin disease classes.
2. Training time and computational requirements.
3. Ensuring good accuracy for rare classes.
4. Integrating ML backend with React frontend.

Solutions:

- Applied data augmentation techniques.
- Used transfer learning with pretrained CNN models.
- Applied weighted loss functions to handle imbalance.
- Developed REST API for seamless frontend-backend communication.

5. Conclusion

The project demonstrates that deep learning models can effectively classify skin diseases from dermoscopic images. With the integration of CNNs and transfer learning, the system achieves high accuracy. The web interface allows easy use for non-technical users, making it a potential assistive tool for dermatologists.

6. Project Architecture

[Insert system architecture diagram here: User → React frontend → Flask API → CNN Model → Results]

7. Future Scope

- Adding more diverse datasets for higher accuracy.
- Deploying as a cloud-based solution.
- Developing a mobile app version.
- Integrating with Electronic Health Records (EHR).
- Using multimodal data (images + patient info) for prediction.